HAMILTONIAN FIXED POINTS BELOW THE STABLE MORSE NUMBER IN DIMENSION TEN, AND AN OBSTRUCTION IN DIMENSION SIX

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Abstract. We construct a simply connected closed Kähler manifold M of real dimension ten and a Hamiltonian diffeomorphism of M with only nondegenerate fixed points whose number is $\lambda(M) - 4 = \mathrm{SM}(M) - 4$, where λ is the integral Morse count and SM is the stable Morse number. The count equals the cyclic integral Floer bound of Bai–Xu, which is therefore sharp already in complex dimension five. The new ingredients are a simply connected threefold F with Brauer-type 2-torsion, obtained as a resolved $\mathbf{Z}/5$-quotient of an invariant Artin–Mumford double solid, a divisor $S \subset F$ with $\pi_1(S) = \mathbf{Z}/5$, and a blow-up along a center that meets the fixed locus of the Hamiltonian involution transversally, so that no new fixed components appear. In the other direction, the Bai–Xu bound excludes simply connected counterexamples in real dimension six, and in real dimension eight every construction of the type of Proposition 2.6 (a finite-order element of a Hamiltonian torus action plus a small invariant perturbation) yields at least $\lambda(M)$ fixed points. Among simply connected closed symplectic manifolds of real dimension at least six, the minimal dimension of a counterexample to the stable Morse bound is therefore 8 or 10.

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1. Introduction

1.1. **Stable Morse number and Hamiltonian fixed points.** Let W be a connected closed smooth manifold. Its *stable Morse number* $\mathrm{SM}(W)$ is the minimum of $\#\,\mathrm{Crit}(\Phi)$ over $k \geq 0$ and Morse functions $\Phi\colon W \times \mathbf{R}^k \to \mathbf{R}$ that coincide with a nondegenerate quadratic form in the $\mathbf{R}^k$-variables outside a compact set; the case $k = 0$ is the ordinary Morse number $\mathrm{Morse}(W) \geq \mathrm{SM}(W)$. For a closed symplectic manifold (M, ω) and a one-periodic Hamiltonian H all of whose fixed points are nondegenerate, the *stable Morse formulation* of the Arnold problem asks whether

$$\#\,\mathrm{Fix}_0(\phi_H^1; H) \;\geq\; \mathrm{SM}(M), \tag{1.1}$$

where Fix_0 denotes fixed points with contractible orbits; see [Gol20, Conjecture 1.2]. Positive results are known under asphericity hypotheses [DG17]. Counterexamples on simply connected Kähler manifolds were found in [OpenAIStable; OpenAI22; OpenAICompanion], the smallest previously recorded fixed dimension being real dimension 22 [OpenAI22]; a non-simply-connected counterexample to the ordinary Morse-number bound in real dimension 12 is given in [OpenAISept23].

For a closed manifold W put

$$b(W) = \sum_i \mathrm{rank}\, H_i(W; \mathbf{Z}), \qquad t_i(W) = d(\mathrm{Tor}\, H_i(W; \mathbf{Z})), \qquad \lambda(W) = b(W) + 2\sum_i t_i(W),$$

where d is the minimal number of generators. Then $\mathrm{SM}(W) \geq \lambda(W)$ for every closed W [OpenAI22, Prop. 2.2], with equality for simply connected W of dimension ≥ 6 by Smale's theorem (§2.2).

1.2. **Main results.** The following theorem lowers the dimension of a simply connected counterexample from 22 to 10.

Theorem A. *There exist a simply connected closed Kähler manifold (M, ω) of complex dimension 5 and a smooth one-periodic Hamiltonian H on M such that every fixed point of ϕ_H^1 is nondegenerate and*

$$\#\,\mathrm{Fix}(\phi_H^1) = \#\,\mathrm{Fix}_0(\phi_H^1; H) = \lambda(M) - 4 = \mathrm{SM}(M) - 4 = \mathrm{Morse}(M) - 4.$$

Moreover the minimal Chern number of M is 1 and $\#\,\mathrm{Fix}(\phi_H^1)$ equals the cyclic integral Floer bound $\lambda_1(M)$ of Bai–Xu (Definition 2.1); this bound is therefore attained in complex dimension 5.

The gap 4 does not depend on any free Betti number. For the resolution chosen in §4 we obtain $\mathrm{SM}(M) = 524$ and $\#\,\mathrm{Fix} = 520$ (Remark 5.5).

Theorem B. *Let (M,ω) be a closed symplectic manifold with a Hamiltonian action of a torus T, and let $g \in T$ have finite order. Suppose that every connected component $C_1,\dots,C_r$ of $\mathrm{Fix}(g)$ has real dimension at most 6 (this holds whenever $\dim_{\mathbf{R}} M = 8$ and g does not act trivially). Then*

$$\sum_{j=1}^{r} \lambda(C_j) \;\geq\; \lambda(M).$$

Consequently, if $\dim_{\mathbf{R}} M = 8$, every Hamiltonian diffeomorphism $\psi^{\epsilon} \circ g$ produced by Proposition 2.6 has at least $\lambda(M)$ nondegenerate fixed points, and at least $\mathrm{SM}(M)$ if M is simply connected.

Theorem B is complemented by necessary conditions valid for every simply connected eight-dimensional counterexample (Proposition 6.1): the minimal Chern number must be 1, and the dominant primes of $\mathrm{Tor}\, H_2(M;\mathbf{Z})$ and $\mathrm{Tor}\, H_3(M;\mathbf{Z})$ must differ. All previously known simply connected constructions [OpenAIStable; OpenAI22; OpenAICompanion] and the construction of Theorem A are of the type excluded by Theorem B in dimension 8. In dimension 6 the Bai–Xu bound already excludes simply connected counterexamples (Proposition 3.1); real dimensions 2 and 4 are discussed in §3. In summary:

Corollary 1.1. *Among simply connected closed symplectic manifolds of real dimension at least 6, the minimal dimension of a counterexample to (1.1) is 8 or 10.*

real dim	status	reference
2	no counterexample	§3
4	simply connected: open, reduces to SM vs. λ	§3
6	no simply connected counterexample	Prop. 3.1
8	open; excluded within the finite-order/torus mechanism	Theorem B, Prop. 6.1
10	simply connected Kähler counterexample, Floer bound sharp	Theorem A
12	non-simply-connected counterexample to the Morse-number bound	[OpenAISept23]
12, 16	simply connected Kähler counterexamples	Remarks 7.1, 7.2
22	simply connected Kähler family, unbounded deficit	[OpenAI22]

TABLE 1. Current status by dimension (nondegenerate Hamiltonians).

1.3. **Previous work versus our contributions.** *Previous work.* The mechanism of [OpenAIStable; OpenAI22; OpenAICompanion] takes $M = X \times Y$, where X is a simply connected Kähler manifold with torsion at two primes and Y is $\mathbf{CP}^1 \times \mathbf{CP}^1$ blown up at four torus-fixed points; the half-turn g_Y of a Hamiltonian circle action on Y fixes four lines, and a small invariant perturbation of $\mathrm{id}_X \times g_Y$ has $4\lambda(X \times \mathbf{CP}^1)$ fixed points. Because d takes the *maximum* over primes in each degree, convolution of the torsion profile of X with the free profiles $(1,1)$ of $\mathbf{CP}^1$ and $(1,6,1)$ of Y can make $\lambda(X \times Y) > 4\lambda(X \times \mathbf{CP}^1)$. The torsion of X comes from blow-ups of a free ambient space along bielliptic surfaces and their products, and the resulting examples live in real dimension 22 [OpenAI22] (and 3332 in [OpenAICompanion], where the cyclic Floer bound is attained). Independently, [OpenAISept23, Thm. 1.1] gives a nondegenerate counterexample to the *ordinary* Morse-number bound on a closed symplectic manifold of real dimension 12 with fundamental group A_5^k (not simply connected), with $\#\,\mathrm{Fix} = \chi(M) + 128 < \mathrm{Morse}(M)$; there the effect comes from π_1, not from torsion. A degenerate six-dimensional example on a quadric with three fixed points, for the critical-point count, is in [OpenAIDeg]. The stable Morse bound is formulated in [Gol20]; Bai–Xu [BX22; BX26] prove the cyclic integral lower bound. Ma's preprint [Ma13, Thm. 1.5] asserts that the minimal number of fixed points equals the Morse number; see §7.2.

Where the previous method fails. The product mechanism cannot work below real dimension 12 (Proposition 6.3): for a simply connected X of real dimension ≤ 6 the torsion of X is a single group placed in degrees $2, 3$, and no exchange of dominant prime can occur. Moreover blow-ups of a free ambient space never create torsion in degree 2, which is forced in dimension 8 by Proposition 6.1.

Our contributions.

(1) A new torsion source: a simply connected threefold F with $\operatorname{Tor} H_2(F) = \operatorname{Tor} H_3(F) = \mathbf{Z}/2$ and *no* 5-torsion, containing a smooth divisor S with $\pi_1(S) = \mathbf{Z}/5$ (§4). The proof that the 5-torsion vanishes uses a permutation-module structure of H_2 of an Artin–Mumford double solid and Lefschetz duality.

(2) A non-product construction: the blow-up of $F \times Y$ along $W = S \times (L_0 \sqcup L_2)$, where $L_0, L_2 \subset Y$ are invariant lines meeting the fixed curves transversally. The transverse center lemma (Lemma 5.1) shows that this blow-up adds torsion to the fixed components only in a controlled way and creates no new fixed components (§5). This gives Theorem A: the first simply connected example below the stable Morse number in real dimension < 22, and a counterexample to the Morse-number assertion of [Ma13] on a simply connected Kähler manifold of real dimension 10.

(3) In dimension 8, necessary conditions together with the unconditional mechanism obstruction Theorem B, based on perfectness of moment maps over every prime field (Lemma 6.4, a specialization of the Borel–Smith inequality; we do not claim novelty for it).

(4) A heuristic explanation of why the transverse-center construction needs real dimension at least 10 (§6.5), and a machine classification of admissible torsion configurations corroborating Theorem B (Appendix A).

The low-dimensional statements of §3 are straightforward consequences of Bai–Xu and duality. The Kähler examples in dimensions 12 and 16 recorded in §7.1 are rearrangements of the same ideas and are not claimed as separate results.

1.4. **Conventions.** Homology has integer coefficients unless indicated. For a finite abelian group A and a prime p, write $d_p(A) = \dim_{\mathbf{F}_p}(A \otimes \mathbf{F}_p)$; then $d(A) = \max_p d_p(A)$. For a space X set $t_i^p(X) = d_p(\operatorname{Tor} H_i(X))$ and $r_i(X) = \operatorname{rank} H_i(X)$. Homology shifted by $2k$ is written $H(\cdot)[2k]$.

2. Torsion counts and fixed-point bounds

2.1. **Degreewise and cyclic generator counts.**

Definition 2.1. For a closed manifold W and $N \in \{1, 2, \dots\} \cup \{\infty\}$ put

$$\lambda_N(W) = b(W) + 2 \sum_{c \in \mathbf{Z}/2N} d\Big(\bigoplus_{i \equiv c \ (2N)} \operatorname{Tor} H_i(W) \Big),$$

with $\mathbf{Z}/2\infty = \mathbf{Z}$, so that $\lambda_\infty = \lambda$.

Since $d(A \oplus B) = \max_p(d_p(A) + d_p(B)) \leq d(A) + d(B)$, one has $\lambda_N \leq \lambda$, with equality as soon as each residue class mod $2N$ contains at most one degree carrying torsion. For finite groups A, B,

$$d(A) + d(B) - d(A \oplus B) > 0 \iff \operatorname{argmax}_p d_p(A) \cap \operatorname{argmax}_p d_p(B) = \varnothing \text{ and } A, B \neq 0. \tag{2.1}$$

We say that A and B have *different dominant primes* in this case.

2.2. **Morse-theoretic bounds.**

Proposition 2.2 ([OpenAI22, Prop. 2.2])**.** *For every connected closed smooth manifold W, $\mathrm{SM}(W) \geq \lambda(W)$; in particular every Morse function on W has at least $\lambda(W)$ critical points.*

Theorem 2.3 (Smale)**.** *Let W be a simply connected closed smooth manifold of dimension $D \geq 6$. Then W admits a Morse function with exactly $r_i(W) + t_i(W) + t_{i-1}(W)$ critical points of index i; hence* $\mathrm{SM}(W) = \mathrm{Morse}(W) = \lambda(W)$.

For $D \geq 9$ this is [OpenAI22, Thm. 2.4] with the handle proof of its appendix, based on [Sma63]. For all $D \geq 6$ it is [Sma62, Thm. 6.1].[1]

2.3. **The cyclic integral Floer bound.** Let N_M be the minimal Chern number of (M, ω): the positive generator of $c_1(TM)(\pi_2(M)) \subset \mathbf{Z}$, or ∞ if this subgroup is zero (Bai–Xu write $N = 0$ in this case).

Theorem 2.4 (Bai–Xu [BX22, Thm. A], see also [BX26, Cor. 1.2])**.** *Let (M, ω) be any closed symplectic manifold and H a one-periodic Hamiltonian with only nondegenerate fixed points. Then*

$$\#\operatorname{Fix}_0(\phi_H^1; H) \;\geq\; \operatorname{rank} H_*(M; \mathbf{Q}) + 2 \sum_{i \in \mathbf{Z}/2N_M} \tau_i^{(2N_M)}(M) = \lambda_{N_M}(M),$$

where $\tau_i^{(2N)}(M)$ is the number of nontrivial invariant factors of the torsion of $\bigoplus_{k \equiv i \ (2N)} H_k(M; \mathbf{Z})$, that is, its minimal number of generators.

Remark 2.5. Rezchikov [Rez22, Thm. 1] phrases an integral lower bound as the minimal rank of a cochain complex of free $\mathbf{Z}$-modules with cohomology $H^*(M; \mathbf{Z})$. His Floer complexes are built from cappings modulo the symplectic area only, and he explicitly avoids the integrally graded variant ([Rez22, §2.1, Remark 2]: "This allows one to define integrally graded Floer groups, which we will avoid in this paper"), so the grading is cyclic and the bound obtained is the cyclic one, consistent with Theorem 2.4. Read with a $\mathbf{Z}$-grading, the statement would give $\#\operatorname{Fix} \geq \lambda(M)$, which Theorem A contradicts.

2.4. **Perturbation of finite-order Hamiltonian symmetries.** The following is [OpenAI22, Prop. 4.2] (Prop. dyn:involution there) with the involution replaced by an element of finite order.

Proposition 2.6. *Let (P, ω) be a closed symplectic manifold and g a Hamiltonian diffeomorphism with $g^k = \mathrm{id}$ for some $k \geq 1$. Let $\operatorname{Fix}(g) = C_1 \sqcup \cdots \sqcup C_r$ and let $h_j \colon C_j \to \mathbf{R}$ be Morse functions. There is a smooth g-invariant $f \colon P \to \mathbf{R}$ with $f|_{C_j} = h_j$ such that for all sufficiently small $\epsilon > 0$ the Hamiltonian flow ψ^s of f satisfies*

$$\operatorname{Fix}(\psi^\epsilon \circ g) = \bigsqcup_j \operatorname{Crit}(h_j),$$

every such fixed point is nondegenerate, and $\psi^\epsilon \circ g$ is the time-one map of a smooth one-periodic Hamiltonian. If P is simply connected, all fixed-point orbits are contractible.

Proof. We indicate the changes to the proof of [OpenAI22, Prop. 4.2]. Average a metric over $\langle g \rangle$; the fixed set is a union of closed submanifolds with $T_x C_j = E_x^+ = \ker(d_x g - I)$. Let $E_x^{\neq}$ be the sum of the (real parts of the) generalized eigenspaces of $d_x g$ for eigenvalues $\neq 1$; $d_x g$ is semisimple of finite order, so $T_x P = E_x^+ \oplus E_x^{\neq}$ and $E_x^{\neq} = \operatorname{im}(d_x g - I)$. If $u \in E_x^+$ and $v = d_x g(w) - w$, then $\omega(u, v) = \omega(d_x g\, u, d_x g\, w) - \omega(u, w) = 0$, so the summands are ω-orthogonal and C_j is symplectic. Let $f = \frac{1}{k} \sum_{m=0}^{k-1} \widetilde{f} \circ g^m$ for an extension $\widetilde{f}$ of the h_j. Then f is g-invariant, $d_x f$ vanishes on $\operatorname{im}(d_x g - I) = E_x^{\neq}$ for $x \in \operatorname{Fix}(g)$, hence X_f is tangent to C_j and equals X_{h_j} there, and g commutes with the flow ψ^s of X_f. Put $\Phi = \psi^\epsilon \circ g$; then $\Phi^k = \psi^{k\epsilon}$. If $\Phi(x) = x$ and $k\epsilon$ is smaller than the constant of [OpenAI22, Lemma 4.1] (short-period lemma), then $X_f(x) = 0$, so $\psi^\epsilon(x) = x$ and $g(x) = x$, and $x \in \operatorname{Crit}(h_j)$. At such x, with $A = d_x X_f$ preserving E_x^+ and $E_x^{\neq}$, the tangent block of $d_x \Phi - I$ is $\exp(\epsilon A^+) - I = \epsilon A^+ + O(\epsilon^2)$ with A^+ invertible by the Morse condition, and the

[1] The torsion-inclusive statement of [Sma62, Thm. 6.1] for $n \geq 6$ was confirmed through secondary sources (Encyclopedia of Mathematics, "Morse inequalities"; arXiv:2607.20259); the original wording was not consulted.

normal block is $d_x g|_{E^{\neq}} \exp(\epsilon A^{\neq}) - I \to d_x g|_{E^{\neq}} - I$, which is invertible. The concatenation argument producing a one-periodic Hamiltonian is unchanged. □

2.5. **Integral blow-up formula.**

Proposition 2.7 ([DL09, Thm. 4.1], [OpenAI22, Prop. 3.3])**.** *Let A be a compact connected Kähler manifold and $Z \subset A$ a closed connected complex submanifold of complex codimension $c \geq 2$. Then $\mathrm{Bl}_Z A$ is compact Kähler, simply connected if A is, and*

$$H_i(\mathrm{Bl}_Z A; \mathbf{Z}) \cong H_i(A; \mathbf{Z}) \oplus \bigoplus_{j=1}^{c-1} H_{i-2j}(Z; \mathbf{Z}).$$

3. Low dimensions

The statements of this section are direct consequences of Theorem 2.4 and Poincaré duality; we record them to delimit Corollary 1.1.

Real dimension 2. For a closed oriented surface Σ_g, SM = Morse = $\lambda = 2g + 2$ (Proposition 2.2 and the standard Morse function). If $g = 0$, $N = 2$ and $\lambda_2(S^2) = 2$; if $g \geq 1$, $\pi_2 = 0$, $N = \infty$ and $\lambda_\infty = \lambda$. Theorem 2.4 gives $\#\operatorname{Fix}_0 \geq 2g + 2 = \mathrm{SM}(\Sigma_g)$, so there is no counterexample.

Simply connected real dimension 4. The homology is torsion-free, so Theorem 2.4 gives $\#\operatorname{Fix} \geq b(M) = \lambda(M)$. A simply connected four-dimensional counterexample would therefore require $\mathrm{SM}(M) > \lambda(M)$. Smale's theorem does not apply in dimension 4, and we do not settle whether SM = λ for simply connected closed 4-manifolds here.

Proposition 3.1. *Let (M, ω) be a simply connected closed symplectic manifold of real dimension 6. Every one-periodic Hamiltonian with only nondegenerate fixed points satisfies $\#\operatorname{Fix}(\phi_H^1) \geq \lambda(M) = \mathrm{SM}(M)$.*

Proof. Write $T_i = \operatorname{Tor} H_i(M)$. By universal coefficients and Poincaré duality, $T_i \cong \operatorname{Tor} H^{i+1}(M) \cong T_{5-i}$. Since H_0 is free and $H_1 = 0$, we get $T_0 = T_1 = T_4 = T_5 = 0$, $T_6 = 0$, and $T_2 \cong T_3$. The two degrees 2 and 3 have different parities, hence lie in different residue classes modulo $2N$ for every $N \in \{1, 2, \ldots, \infty\}$. Therefore $\lambda_N(M) = \lambda(M)$ for every N. Theorem 2.4 gives $\#\operatorname{Fix} \geq \lambda(M)$, and Theorem 2.3 with $D = 6$ gives $\lambda(M) = \mathrm{SM}(M)$; contractibility is automatic. □

Remark 3.2. Proposition 3.1 concerns nondegenerate Hamiltonians. The six-dimensional example of [OpenAIDeg], a Hamiltonian diffeomorphism of the quadric Q^3 with three fixed points, has a degenerate fixed point and concerns the critical-point count; it does not conflict with Proposition 3.1.

4. An Artin–Mumford quotient threefold

4.1. **An invariant quartic symmetroid web.** Let $\zeta = e^{2\pi i/5}$ and let $G = \langle g \rangle \cong \mathbf{Z}/5$ act on $\mathbf{P}^3$ by

$$g \cdot [x_0 : x_1 : x_2 : x_3] = [x_0 : \zeta^4 x_1 : \zeta^3 x_2 : \zeta x_3].$$

Consider the web of quadrics given by the symmetric matrix of linear forms

$$A(x) = \begin{pmatrix} 6x_0 & 5x_1 & 4x_2 & 0 \\ 5x_1 & 8x_2 & 0 & 4x_3 \\ 4x_2 & 0 & 7x_3 & 6x_0 \\ 0 & 4x_3 & 6x_0 & 5x_1 \end{pmatrix},$$

and let $B = \{f = 0\}$, $f = \det A$:

$$f = 900x_0^2x_1^2 - 1728x_0^3x_2 - 640x_1x_2^3 - 875x_1^3x_3 + 720x_0x_1x_2x_3 + 256x_2^2x_3^2 - 672x_0x_3^3.$$

Every monomial of f has g-weight 3 mod 5, so $f(gx) = \zeta^3 f(x)$.

Lemma 4.1 (machine-verified, Appendix B)**.** (1) *The four matrices $A_0, \dots, A_3$ (coefficients of $x_0, \dots, x_3$) are linearly independent, the web has no base points, and no member has rank ≤ 1.*

(2) *f is irreducible over $\mathbf{Q}$, the singular scheme of B equals the rank-≤ 2 scheme of A, and it is reduced of length 10. Thus B has exactly ten singular points, all ordinary double points (A_1). Since $\mathrm{Sing}(B)$ is finite and a reducible surface is singular along a curve, f is absolutely irreducible. The only g-fixed points of $\mathbf{P}^3$ are the coordinate points, which are smooth points of B (item 3) and $|G| = 5$ is prime, so the nodes form two free G-orbits.*

(3) *The four coordinate points $p_0, \dots, p_3$ lie on B and are smooth points of B.*

(4) *(Beauville's condition (ii).) If a line of $\mathbf{P}^3$ is contained in the singular locus of a quadric of the web, it is not contained in any other quadric of the web. Equivalently, at each of the ten rank-2 points the three quadratic conditions on the web imposed by the kernel line are independent; this is checked exactly over $\overline{\mathbf{Q}}$.*

(5) *The ten nodes impose independent conditions on quadrics: the Hilbert function of $\mathbf{C}[x_0, \dots, x_3]/I_{\mathrm{nodes}}$ is $1, 4, 10, 10, \dots$, so $H^0(\mathbf{P}^3, I_{\mathrm{nodes}}(2)) = 0$.*

Let $\overline{X} \to \mathbf{P}^3$ be the double cover $w^2 = f$ (a hypersurface in $\mathbf{P}(1,1,1,1,2)$) and $X := \mathrm{AM}_5$ its blow-up at the ten nodes. The lift $\hat{g}\colon (x, w) \mapsto (gx, \zeta^4 w)$ has order 5 (note $\zeta^8 = \zeta^3$) and lifts to X; we denote it again by g.

Lemma 4.2. *g acts on X with exactly four fixed points, one over each p_k, each of type $\frac{1}{5}(1,1,4)$. All other points have trivial stabilizer.*

Proof. The weights $0, 4, 3, 1$ are pairwise distinct, so $\mathrm{Fix}(g|_{\mathbf{P}^3}) = \{p_0, \dots, p_3\}$; the nodes form free orbits, so no exceptional quadric contains a fixed point. Since $p_k \in B$, the fiber over p_k is the single point $w = 0$. At p_0 use affine coordinates $u_k = x_k/x_0$ of weights $4, 3, 1$. The linear part of $f(1, u)$ is a nonzero multiple of u_2 ($\partial f/\partial x_2(p_0) = -1728$); hence (w, u_1, u_3) are local coordinates on X, of weights $(4, 4, 1)$, i.e. type $\frac{1}{5}(1,1,4)$ after multiplying by 4. At p_1, p_2, p_3 the nonvanishing partials are $\partial_{x_3} f(p_1) = -875$, $\partial_{x_1} f(p_2) = -640$, $\partial_{x_0} f(p_3) = -672$, and the same computation gives weights $(1,1,4)$, $(3,2,3)$, $(2,3,2)$, all of type $\frac{1}{5}(1,1,4)$. As $|G|$ is prime, every non-fixed point has trivial stabilizer. □

4.2. **Topology of the invariant double solid.**

Proposition 4.3. *$X = \mathrm{AM}_5$ is simply connected and*

$$H_*(X; \mathbf{Z}) = (\mathbf{Z},\ 0,\ \mathbf{Z}^{11} \oplus \mathbf{Z}/2,\ \mathbf{Z}/2,\ \mathbf{Z}^{11},\ 0,\ \mathbf{Z}) \quad \textit{in degrees } 0, \dots, 6.$$

Proof. Betti numbers. By Lemma 4.1(5) the ten nodes impose independent conditions on quadrics. For a double solid branched along a surface of degree d whose only singularities are μ nodes, Clemens' defect δ is the number of dependent conditions imposed by the nodes on forms of degree $\frac{3}{2}d - 4$, and the big resolution $\widetilde{Y}$ (blow-up of the nodes) has

$$h^{1,1}(\widetilde{Y}) = 1 + \mu + \delta, \qquad h^{1,2}(\widetilde{Y}) = \tbinom{3d/2-1}{3} - 4\tbinom{d/2}{3} - \mu + \delta$$

[Cle83]; we use the formulation of [Cyn07, §1 and Cor. 5.1], which states these formulas verbatim (we have not consulted the page of [Cle83] where they appear). Here $d = 4$, $\mu = 10$, and the forms have degree 2, so $\delta = 0$ and $h^{1,1}(X) = 11$, $h^{1,2}(X) = 10 - 0 - 10 + 0 = 0$. Since $h^{3,0} = h^{2,0} = 0$ for a quartic double solid, $b_3(X) = 0$ and $b_2(X) = 11$. This agrees with the Euler characteristic: a quartic with ten nodes has $\chi = 24 - 10 = 14$, so $\chi(\overline{X}) = 2\chi(\mathbf{P}^3) - 14 = -6$, and replacing each node by a quadric adds 3, giving $\chi(X) = 24 = 2 + 2 \cdot 11$.

Nonzero 2-torsion and simple connectivity. The web is base point free and satisfies Beauville's condition (ii) (Lemma 4.1(1),(4)). These are exactly the hypotheses of [Bea15, §6.3]. Under them Beauville shows that the discriminant has ten ordinary double points and no other singularities, and [Bea15, Thm. 6 and its proof] shows that the blown-up double solid X is unirational and that $\mathrm{Tor}\, H^3(X; \mathbf{Z})$ contains a nonzero element of order 2. Unirational smooth projective varieties are simply connected (Serre), so $\pi_1(X) = 1$.

Exactness, by deformation. Beauville's argument gives $\operatorname{Tor} H^3 \neq 0$, and Voisin [Voi15, proof of Cor. 2.6] observes that every nodal quartic double solid has a unirational parametrization of degree 2, so the torsion is killed by 2. We could not find a source stating that $\operatorname{Tor} H^3 \cong \mathbf{Z}/2$ (rather than $(\mathbf{Z}/2)^k$) for *every* web satisfying (i) and (ii). Endraß [End99, Lemma 1.2] embeds the torsion into a group of even sets of nodes, but we have not computed that group for our web. We therefore deform to the Artin–Mumford example, for which $\operatorname{Tor} H^3 \cong \mathbf{Z}/2$ [AM72], [Del73, Constr. 2.2]. Since $\operatorname{Tor} H^3$ is a birational invariant of smooth projective threefolds [Del73, Prop. 2.1], this holds for every smooth projective model of that example, in particular for the blown-up double solid of the corresponding symmetroid ([Bea15, §6.3]). [**TODO:** confirm that the Artin–Mumford example of [Del73] is birational to a double solid branched along a quartic symmetroid satisfying (i) and (ii); [Bea15, §6.3] and Cossec state that the Artin–Mumford example is such a double solid.]

Let $\mathrm{Gr} = \mathrm{Gr}(4, \operatorname{Sym}^2 \mathbf{C}^4)$ (irreducible) and let $\mathcal{D} \subset \mathbf{P}^3 \times \mathrm{Gr}$ be the relative rank-≤ 2 scheme and $\mathcal{S} \supset \mathcal{D}$ the relative singular scheme of the symmetroids; $\mathcal{S}$ is proper over Gr. The partial derivatives of $\det A_\Lambda$ are combinations of 3×3 minors, so $\mathcal{D}_\Lambda \subset \mathcal{S}_\Lambda$. The variety $\Sigma_2 \subset \mathbf{P}^9$ of symmetric 4×4 matrices of rank ≤ 2 is irreducible, arithmetically Cohen–Macaulay, of codimension 3 and degree 10; the degree is the case $n = 4$, $r = 2$ of the Harris–Tu formula $\prod_{j=0}^{n-r-1} \binom{n+j}{n-r-j} / \binom{2j+1}{j}$ [HT84] (we use the statement as quoted in [Ble+12, §1]). When $\mathcal{D}_\Lambda$ is finite, it is a proper linear section of the Cohen–Macaulay variety Σ_2, so its length equals $\deg \Sigma_2 = 10$. Hence $\operatorname{length} \mathcal{S}_\Lambda \geq 10$ whenever it is finite. On the open locus over which $\mathcal{S}$ is quasi-finite, the fibre length is upper semicontinuous, and the étale locus of a finite flat family is open; hence

$$U_0 = \{\Lambda : \mathcal{S}_\Lambda \text{ is reduced of length } 10\}$$

is Zariski open in Gr. It contains every web satisfying (i) and (ii) (by Beauville's statement above), in particular the Artin–Mumford web and our web (Lemma 4.1(2)). Being a nonempty Zariski open subset of an irreducible variety, it is connected. Over U_0 the nodes form a finite étale multisection, and by the holomorphic Morse lemma with parameters the double solids are locally analytically of the form $w^2 = y_1^2 + y_2^2 + y_3^2$ times the base near the nodes. Blowing up the multisection therefore gives a smooth proper family, and by Ehresmann's theorem all members are diffeomorphic. Therefore $\operatorname{Tor} H^3(X; \mathbf{Z}) \cong \mathbf{Z}/2$. Universal coefficients and duality give $\operatorname{Tor} H_2 \cong \operatorname{Tor} H^3 = \mathbf{Z}/2$ and $H_3 \cong H^3 = \mathbf{Z}/2$. □

4.3. **Quotient, resolution, fundamental group.** Let $X^\circ \subset X$ be the complement of small G-invariant open balls around the four fixed points, $U = X^\circ/G$, and $\partial U = \bigsqcup_{k=0}^{3} L_k$ with $L_k \cong S^5/(\mathbf{Z}/5)$ lens spaces. Each fixed point has type $\frac{1}{5}(1,1,4)$; let $\rho_k \colon R_k \to \mathbf{C}^3/G$ be the toric resolution given by iterated star subdivision of the cone of $\frac{1}{5}(1,1,4)$ at the lattice points $\frac{1}{5}(1,1,4)$, $\frac{1}{5}(2,2,3)$, $\frac{1}{5}(3,3,2)$, $\frac{1}{5}(4,4,1)$ (Appendix B). It has four exceptional divisors and nine maximal cones; it is a composition of weighted blow-ups, hence projective. Define

$$F = U \cup_{\partial U} \textstyle\bigsqcup_k R_k = \text{the resolution of } X/G \text{ at its four singular points.}$$

F is a smooth projective threefold. Each R_k, a small neighbourhood of the exceptional set of ρ_k, deformation retracts onto the exceptional set $E = D_1 \cup D_2 \cup D_3 \cup D_4$, where D_i is the toric divisor of the interior ray $v_i = \frac{i}{5}(1,1,4) \bmod \mathbf{Z}^3$. In the fan (computed in `computations/sage/bF2.sage`) the maximal cones are $\langle e_1, e_2, v_4\rangle$, $\langle e_3, e_1, v_1\rangle$, $\langle e_3, e_2, v_1\rangle$ and $\langle e_a, v_i, v_{i+1}\rangle$ ($a = 1, 2$, $i = 1, 2, 3$). Hence each D_i is a smooth complete toric surface, $D_i \cap D_j = \varnothing$ for $|i - j| \geq 2$, there are no triple intersections, and $D_i \cap D_{i+1}$ is the invariant curve $V(\langle v_i, v_{i+1}\rangle) \cong \mathbf{CP}^1$. A smooth complete toric surface has free homology concentrated in even degrees [Ful93, §5.2, pp. 101–103]. An invariant curve on it is primitive in H_2, since it meets an adjacent invariant curve with intersection number 1. Mayer–Vietoris along the chain $D_1 \cup D_2 \cup D_3 \cup D_4$ therefore shows that $H_*(R_k) \cong H_*(E)$ is free and concentrated in even degrees. Its Betti numbers are $(1, 4, 4)$ (the Euler characteristic 9 is the number of maximal cones).

Lemma 4.4. $\pi_1(F) = 1$.

Proof. X° is simply connected (removing points from a simply connected 6-manifold), so $\pi_1(U) = G$ since G acts freely on X°. A loop in L_k lifting to a path from x to gx in the boundary sphere represents g, so $\pi_1(L_k) \to \pi_1(U)$ is onto. The R_k are simply connected. Van Kampen gives $\pi_1(F) = 1$. (Equivalently, by Armstrong's theorem [Arm68] $\pi_1(X/G) = G/\langle\text{elements with fixed points}\rangle = 1$, and resolving isolated quotient singularities with simply connected exceptional sets does not change π_1.) □

4.4. **Integral homology of F.**

Proposition 4.5. *$H_1(F) = H_5(F) = 0$, $b_1 = b_3 = b_5 = 0$, $b_2(F) = b_4(F) = 19$, $b(F) = 40$, and $\operatorname{Tor} H_i(F) \cong \operatorname{Tor} H_{5-i}(F)$; in particular $\operatorname{Tor} H_4(F) = 0$.*

Proof. By the Lefschetz formula $\chi(X/G) = \frac{1}{5}(\chi(X) + 4 \cdot \#\operatorname{Fix}(g)) = \frac{1}{5}(24 + 16) = 8$. Each resolution replaces a point by a set of Euler characteristic 9, so $\chi(F) = 8 + 4 \cdot 8 = 40$. Rationally $H_*(U;\mathbf{Q}) = H_*(X^\circ;\mathbf{Q})^G$ and $H_{\text{odd}}(X;\mathbf{Q}) = 0$; with the even homology of the R_k, Mayer–Vietoris gives $b_{\text{odd}}(F) = 0$, hence $b_2 = b_4 = 19$ (the invariant part of $H_2(X;\mathbf{Q})$ has rank 3 by the permutation-module description in Lemma 4.6, and $3 + 4 \cdot 4 = 19$). $H_1(F) = 0$ by Lemma 4.4. F is a closed oriented 6-manifold, so $\operatorname{Tor} H_i \cong \operatorname{Tor} H^{i+1} \cong \operatorname{Tor} H_{5-i}$. □

Lemma 4.6. $\operatorname{Tor} H_2(F) \cong \operatorname{Tor} H_3(F) \cong \mathbf{Z}/2$. *In particular F has no 5-torsion.*

Proof. By Proposition 4.5 it suffices to compute $\operatorname{Tor} H_2(F)$ and $\operatorname{Tor} H_3(F)$ one prime at a time. Mayer–Vietoris for $F = U \cup R$, $R = \bigsqcup R_k$, reads

$$H_i(\partial U) \to H_i(U) \oplus H_i(R) \to H_i(F) \to H_{i-1}(\partial U), \tag{4.1}$$

with $H_*(L_k) = (\mathbf{Z}, \mathbf{Z}/5, 0, \mathbf{Z}/5, 0, \mathbf{Z})$ and $H_*(R)$ free, in even degrees.

Step 1 (input). $\operatorname{Tor} H_2(X) = \mathbf{Z}/2$ and $H_3(X) = \mathbf{Z}/2$ exactly (Proposition 4.3), and $H_i(X^\circ) = H_i(X)$ for $i \le 4$.

Step 2 (localization at $\ell \neq 5$). $H_i(\partial U)_{(\ell)} = 0$ for $i = 1, 2, 3$, so (4.1) gives $H_i(F)_{(\ell)} \cong H_i(U)_{(\ell)} \oplus H_i(R)_{(\ell)}$ for $i = 2, 3$, and $\operatorname{Tor} H_i(F)_{(\ell)} = \operatorname{Tor} H_i(U)_{(\ell)}$.

Step 3 (transfer). As $|G|$ is invertible in $\mathbf{Z}_{(\ell)}$, the Cartan–Leray spectral sequence collapses to $H_i(U;\mathbf{Z}_{(\ell)}) \cong H_i(X^\circ;\mathbf{Z}_{(\ell)})_G \cong H_i(X^\circ;\mathbf{Z}_{(\ell)})^G$ (averaging).

Step 4 (torsion of invariants). For a $\mathbf{Z}_{(\ell)}[G]$-module H with $|G|$ invertible, $H^1(G;\operatorname{Tor} H) = 0$, so $\operatorname{Tor}(H^G) = (\operatorname{Tor} H)^G$. For $\ell = 2$, G acts trivially on $\mathbf{Z}/2$, so $\operatorname{Tor} H_2(F)_{(2)} \cong \operatorname{Tor} H_3(F)_{(2)} \cong \mathbf{Z}/2$. For $\ell \notin \{2, 5\}$ there is no ℓ-torsion.

Step 5 (the prime 5). Let $h \in H^2(X)$ be the pullback of the hyperplane class and $E_1, \dots, E_{10}$ the classes of the exceptional quadrics; g fixes h and permutes the E_i in two free orbits. Pair them with the curve classes $\tilde{\ell}$ (preimage of a general line, $h \cdot \tilde{\ell} = 2$, $E_i \cdot \tilde{\ell} = 0$) and f_i (a ruling of the i-th quadric, $h \cdot f_i = 0$, $E_j \cdot f_i = -\delta_{ij}$, since $E_i|_{E_i} = \mathcal{O}(-1,-1)$). The intersection matrix is $\operatorname{diag}(2, -1, \dots, -1)$, of determinant ± 2. Hence $h, E_1, \dots, E_{10}$ span a sublattice of index dividing 2 in $H^2(X)/\operatorname{Tor}$, and

$$H^2(X;\mathbf{Z}_{(5)}) \cong H_2(X;\mathbf{Z}_{(5)}) \cong \mathbf{Z}_{(5)} \oplus \mathbf{Z}_{(5)}[G]^2$$

as $\mathbf{Z}_{(5)}[G]$-modules (permutation modules are self-dual). In the Cartan–Leray spectral sequence $E^2_{p,q} = H_p(G; H_q(X^\circ;\mathbf{Z}_{(5)})) \Rightarrow H_{p+q}(U;\mathbf{Z}_{(5)})$ [Bro82, p. VII.7] we have $H_1(X^\circ) = 0$, $H_3(X^\circ;\mathbf{Z}_{(5)}) = 0$, and $E^2_{0,2} = (\mathbf{Z} \oplus \mathbf{Z}[G]^2)_G \otimes \mathbf{Z}_{(5)} \cong \mathbf{Z}^3_{(5)}$. Since $H_2(G;\mathbf{Z}) = 0$, the only possible differential into $E_{0,2}$ comes from the finite group $E^3_{3,0} = H_3(G;\mathbf{Z}) = \mathbf{Z}/5$; it is zero because $E^2_{0,2} \cong \mathbf{Z}^3_{(5)}$ is torsion free. Hence $H_2(U;\mathbf{Z}_{(5)}) \cong \mathbf{Z}^3_{(5)}$. Since moreover $b_3(U) = 0$, universal coefficients give $H^3(U;\mathbf{Z}) \otimes \mathbf{Z}_{(5)} = 0$, and Lefschetz duality (U is a compact oriented 6-manifold with boundary) gives $H_3(U, \partial U)_{(5)} \cong H^3(U)_{(5)} = 0$. Hence $H_3(\partial U) \to H_3(U)$ is onto after localizing at 5. As $H_3(R) = 0$ and $H_2(\partial U) = 0$, (4.1) gives $H_3(F)_{(5)} = \operatorname{coker}(H_3(\partial U) \to H_3(U))_{(5)} = 0$.

Step 6 (duality at 5). From (4.1) alone, $H_2(F)_{(5)}$ is an extension of $\ker(H_1(\partial U) \to H_1(U)) \cong (\mathbf{Z}/5)^3$ by the free module $H_2(U)_{(5)} \oplus H_2(R)_{(5)}$, so Mayer–Vietoris does not by itself exclude a $\mathbf{Z}/5$ in $H_2(F)$. Torsion-freeness comes from duality: $\operatorname{Tor} H_2(F)_{(5)} \cong \operatorname{Tor} H_3(F)_{(5)} = 0$ by Step 5. □

Remark 4.7. The exactness of Lemma 4.6 is essential. If $\operatorname{Tor} H_2(F) = (\mathbf{Z}/2)^2$, or if F had 5-torsion in degrees $2, 3$, the gap of Theorem A (and of Remark 7.1) would collapse to ≤ 0 (`computations/homology/w10.py`, `computations/homology/lam.py`). Note that $H_3(U; \mathbf{Z}_{(5)})$ itself has order 25 (contributions $E_{1,2}^{\infty}$ and $E_{3,0}^{\infty}$); it is killed in F by the lens space boundary.

4.5. **A Godeaux-type divisor.** The G-invariant quintic monomials (weight sum $\equiv 0 \bmod 5$) span a 12-dimensional space. Fix the invariant quintic

$$\begin{aligned} Q = {} & 6x_0^5 + 9x_1^5 + 17x_0x_1^3x_2 + 15x_0^2x_1x_2^2 + 13x_2^5 + 14x_0^3x_1x_3 + 10x_1^2x_2^2x_3 \\ & + 10x_0x_2^3x_3 + x_0x_1^2x_3^2 + 19x_0^2x_2x_3^2 + 15x_1x_2x_3^3 + 5x_3^5. \end{aligned}$$

Lemma 4.8 (machine-verified, Appendix B)**.** *The surface $\{Q = 0\}$ is smooth, avoids $p_0, \dots, p_3$ and the ten nodes of B, and meets B transversally in a smooth curve Δ.*

Let $\widetilde{S} \subset X$ be the preimage of $\{Q = 0\}$; it is the double cover of the quintic surface branched along $\Delta \in |4H|$, and lies in the locus where $X \to \overline{X}$ is an isomorphism and G acts freely. Put $S = \widetilde{S}/G \subset F$.

Proposition 4.9. *S is a smooth divisor in F with $\pi_1(S) = \mathbf{Z}/5$; its homology is*

$$H_*(S; \mathbf{Z}) = (\mathbf{Z},\ \mathbf{Z}/5,\ \mathbf{Z}^{40} \oplus \mathbf{Z}/5,\ 0,\ \mathbf{Z}).$$

Proof. Smoothness and the divisor property follow from Lemma 4.8 and the fact that $\widetilde{S}$ avoids the fixed points. Simple connectivity of $\widetilde{S}$: by Nori's theorem [Nor83], since Δ is a smooth curve with $\Delta^2 > 0$ on the simply connected quintic surface, π_1 of the complement of Δ is abelian, hence cyclic generated by a meridian γ; the double cover corresponds to $\langle \gamma^2 \rangle$, which is generated by a meridian of $\widetilde{\Delta}$ and dies in $\widetilde{S}$. Here $H_1(\Sigma \setminus \Delta) = \mathbf{Z}/4$, because the hyperplane class H of the quintic surface Σ is primitive ($H^2 = 5$ is squarefree), so $\Delta \in |4H|$ has divisibility exactly 4. The abelianness statement is [Nor83, Prop. 3.27], applied with $D = \Delta$ (smooth, so $r(\Delta) = 0$ and the hypothesis $\Delta^2 > 2r(\Delta)$ holds) and $E = \varnothing$: the kernel of $\pi_1(\Sigma \setminus \Delta) \to \pi_1(\Sigma) = 1$ is finitely generated abelian. Hence $\pi_1(S) = G = \mathbf{Z}/5$, $H_1(S) = \mathbf{Z}/5$, $\operatorname{Tor} H_2(S) \cong \operatorname{Tor} H_1(S) = \mathbf{Z}/5$ and $H_3(S) = 0$. For Betti numbers, χ of a smooth quintic surface is 55 and Δ has $2g - 2 = \Delta \cdot (\Delta + K) = 4H \cdot 5H = 100$, so $\chi(\widetilde{S}) = 2 \cdot 55 + 100 = 210$, $\chi(S) = 42$ and $b_2(S) = 40$. □

We call S *Godeaux-type* because $\pi_1(S) = \mathbf{Z}/5$; it is not a Godeaux surface (its geometric genus is positive).

Remark 4.10. This gives, in the Kähler category, a simply connected compact Kähler threefold with 2-primary Brauer-type torsion containing a smooth divisor with 5-torsion in H_1. The Brauer group is a birational invariant, which is why one cannot start from a rational threefold such as the toric resolution of $\mathbf{P}^3/\mathbf{Z}_5$.

5. The ten-dimensional construction

5.1. **The surface Y and two invariant lines.** Let Y be $\mathbf{CP}^1 \times \mathbf{CP}^1$ blown up at the four points $\{0, \infty\}^2$, with the diagonal circle action $t \cdot (z, w) = (e^{2\pi i t}z, e^{2\pi i t}w)$ lifted to Y and an invariant Kähler form. By [OpenAI22, Prop. 4.3], the half-turn g_Y is a Hamiltonian involution with $\operatorname{Fix}(g_Y) = E_{00} \sqcup E_{0\infty} \sqcup E_{\infty 0} \sqcup E_{\infty\infty}$ (the exceptional curves), $dg_Y = +1$ on TE and -1 on the normal line; Y is simply connected with $H_*(Y) = (\mathbf{Z}, 0, \mathbf{Z}^6, 0, \mathbf{Z})$; and $c_1(TY)[E] = 1$.

Let $L_0, L_2 \subset Y$ be the proper transforms of $\{0\} \times \mathbf{CP}^1$ and $\{\infty\} \times \mathbf{CP}^1$. They are disjoint invariant smooth rational curves; g_Y rotates each by π, with fixed points ("poles") $L_0 \cap E_{00}$, $L_0 \cap E_{0\infty}$, $L_2 \cap E_{\infty 0}$, $L_2 \cap E_{\infty\infty}$. Thus every fixed curve contains exactly one pole, and at a pole y the curves L and E meet transversally, so $T_yY = T_yE \oplus T_yL$ with $dg_Y = \mathrm{diag}(1, -1)$.

5.2. **Transverse center lemma.**

Lemma 5.1 (transverse center lemma)**.** *Let A be a complex manifold, g a holomorphic automorphism of finite order, $\mathrm{Fix}(g) = \bigsqcup_j C_j$, and $W \subset A$ a g-invariant closed complex submanifold of codimension ≥ 2. Assume that for every $p \in W \cap \mathrm{Fix}(g)$*

$$E_p^{\neq} := \mathrm{im}(d_pg - I) \subset T_pW. \tag{5.1}$$

Let $\hat{g}$ be the lift of g to $\widehat{A} = \mathrm{Bl}_W A$. Then

$$\mathrm{Fix}(\hat{g}) = \bigsqcup_j \widehat{C}_j, \qquad \widehat{C}_j \cong \mathrm{Bl}_{W \cap C_j} C_j,$$

where $\widehat{C}_j$ is the proper transform of C_j (equal to C_j if $W \cap C_j = \varnothing$, and to C_j blown up along a smooth center otherwise). In particular $\hat{g}$ has no fixed components other than the proper transforms of those of g.

Proof. Write $\pi\colon \widehat{A} \to A$ and $E = \pi^{-1}(W) = \mathbf{P}(N_W)$.

(a) Cleanness. $W \cap \mathrm{Fix}(g) = \mathrm{Fix}(g|_W)$ is a closed submanifold of W with $T_p(W \cap C_j) = \ker(d_pg - I) \cap T_pW = T_pC_j \cap T_pW$; so W and each C_j intersect cleanly. By (5.1), $T_pA = T_pC_j \oplus E_p^{\neq} \subset T_pC_j + T_pW$, so $T_pA = T_pC_j + T_pW$.

(b) Fixed points off E, and over non-fixed points of W. π is a g-equivariant isomorphism off E, so $\mathrm{Fix}(\hat{g}) \setminus E = \mathrm{Fix}(g) \setminus W$. If $p \in W$ and $g(p) \neq p$, then $\hat{g}$ maps the fiber E_p to $E_{g(p)} \neq E_p$, so it has no fixed points there.

(c) Fibers over $W \cap \mathrm{Fix}(g)$. For $p \in W \cap C_j$, $\hat{g}$ acts on $E_p = \mathbf{P}(T_pA/T_pW)$ through the map induced by d_pg. By (5.1), T_pA/T_pW is the image of $E_p^{+} = T_pC_j$, on which $d_pg = \mathrm{id}$. Hence $\hat{g}$ fixes E_p pointwise.

(d) Identification. By (a) the natural map $T_pC_j/T_p(W \cap C_j) \to T_pA/T_pW$ is injective (cleanness) and surjective ($T_pA = T_pC_j + T_pW$), so $\mathbf{P}(N_{W\cap C_j/C_j})|_p \to E_p$ is an isomorphism. By cleanness the scheme-theoretic intersection $W \cap C_j$ is the smooth submanifold above, so the strict transform $\widehat{C}_j$ (the closure of $C_j \setminus W$) is isomorphic to $\mathrm{Bl}_{W\cap C_j} C_j$ [Har77, p. II.7.15]; it is a closed connected submanifold whose intersection with E is its exceptional divisor $\mathbf{P}(N_{W\cap C_j/C_j})$, which by the isomorphism just noted is all of $E|_{W\cap C_j}$. It is contained in the closed set $\mathrm{Fix}(\hat{g})$. Combining (b) and (c), as sets

$$\mathrm{Fix}(\hat{g}) = (\mathrm{Fix}(g) \setminus W) \cup E|_{W\cap \mathrm{Fix}(g)} = \bigsqcup_j ((C_j \setminus W) \cup E|_{W\cap C_j}) = \bigsqcup_j \widehat{C}_j.$$

The $\widehat{C}_j$ are disjoint because the C_j are, and $\mathrm{Fix}(\hat{g})$, a closed complex submanifold, is their disjoint union. □

Remark 5.2. Without (5.1) the lemma fails: blowing up $S \times \{y\}$ with y a pole, or $S \times E$, produces additional fixed components (copies of S or $S \times \mathbf{CP}^1$ inside E) that carry torsion and destroy the gap; this is a direct computation with the bookkeeping of `computations/homology/lam.py`. See also §6.5.

5.3. **The manifold M and its fixed locus.** Let F and $S \subset F$ be as in §4 and put

$$W = S \times (L_0 \sqcup L_2) \subset F \times Y, \qquad M = \mathrm{Bl}_W(F \times Y).$$

W has complex dimension 3, hence codimension 2 in the fivefold $F \times Y$.

Proposition 5.3. *M is a simply connected compact Kähler manifold of complex dimension 5 with a Hamiltonian circle action lifting $\mathrm{id}_F \times$ (diagonal action). Its half-turn $\hat{g}$ is a Hamiltonian involution with*

$$\mathrm{Fix}(\hat{g}) = C_1' \sqcup C_2' \sqcup C_3' \sqcup C_4', \qquad C_j' \cong \mathrm{Bl}_{S\times\{\mathrm{pt}\}}(F \times \mathbf{CP}^1),$$

each C_j' simply connected of real dimension 8. The minimal Chern number of M is 1.

Proof. Kählerness and simple connectivity follow from Proposition 2.7. W is invariant, so the holomorphic circle action lifts; averaging a Kähler form and using $H^1(M;\mathbf{R}) = 0$ makes it Hamiltonian, and its half-turn $\hat{g}$ is a Hamiltonian involution lifting $g = \mathrm{id}_F \times g_Y$. We have $\mathrm{Fix}(g) = \bigsqcup_j F \times E_j$ and $W \cap \mathrm{Fix}(g) = S \times \{\text{poles}\}$. At $p = (s, y)$ with y a pole, $E_p^{\neq} = \{0\} \oplus T_y L \subset T_s S \oplus T_y L = T_p W$, so (5.1) holds. Lemma 5.1 gives $\mathrm{Fix}(\hat{g}) = \bigsqcup \widehat{F \times E_j}$ with $\widehat{F \times E_j} \cong \mathrm{Bl}_{S\times\{y_j\}}(F \times E_j)$, y_j the unique pole on E_j. These are simply connected by Proposition 2.7. Finally, for $x \in F \setminus S$ the sphere $\{x\} \times E_{00}$ is disjoint from W and has $c_1 = 1$. □

5.4. **Integral homology and torsion profiles.** By Proposition 2.7 (codimension 2 in both cases),

$$H_*(M) \cong H_*(F \times Y) \oplus 2\, H_*(S \times \mathbf{CP}^1)[2], \qquad H_*(C_j') \cong H_*(F \times \mathbf{CP}^1) \oplus H_*(S)[2]. \tag{5.2}$$

All Künneth factors other than F, S have free homology, so no Tor terms arise. Write $a = b_2(F)$ and $\beta = b_2(S)$.

	M						C_j'			
degree	2	3	4	5	6	7	2	3	4	5
d_2	1	1	6	6	1	1	1	1	1	1
d_5	0	2	2	2	2	0	0	1	1	0
$t_i = \max$	1	2	6	6	2	1	1	1	1	1

TABLE 2. Primary torsion multiplicities $d_p(\mathrm{Tor}\, H_i)$. All torsion is elementary abelian. $\sum_i t_i(M) = 18$, $\sum_i t_i(C_j') = 4$.

degree	0	2	4	6	8	10	total
$r_i(M)$	1	$a+8$	$7a+2\beta+3$	$7a+2\beta+3$	$a+8$	1	$16a+4\beta+24$
$r_i(C_j')$	1	$a+2$	$2a+\beta$	$a+2$	1		$4a+\beta+6$

TABLE 3. Free ranks (all odd ranks vanish). $b(M) = 4b(C_j')$.

Derivation of the tables. F has $\mathrm{Tor}\, H_2 = \mathrm{Tor}\, H_3 = \mathbf{Z}/2$; Künneth with Y (free ranks $1, 6, 1$ in degrees $0, 2, 4$) places 2-torsion of multiplicities $(1, 6, 1)$ in degrees $(2, 4, 6)$ and $(3, 5, 7)$. $S \times \mathbf{CP}^1$ has 5-torsion $\mathbf{Z}/5$ in degrees $1, 2, 3, 4$; shifted by 2 and doubled, this gives $(2, 2, 2, 2)$ in degrees $3, \ldots, 6$. For C_j': $F \times \mathbf{CP}^1$ has 2-torsion in degrees $2, 3, 4, 5$, and $H(S)[2]$ adds $\mathbf{Z}/5$ in degrees $3, 4$. The free ranks follow in the same way from $r(F) = (1, a, a, 1)$, $r(S) = (1, \beta, 1)$. □

Corollary 5.4.

$$\lambda(M) = b(M) + 36, \qquad \lambda(C_j') = \tfrac{1}{4} b(M) + 8, \qquad \lambda(M) - \sum_{j=1}^{4} \lambda(C_j') = 4,$$

independently of a and β. Moreover $\lambda_1(M) = b(M) + 32 = \sum_j \lambda(C_j')$.

Proof. The first three identities follow from the tables. For λ_1, the even-degree torsion of M has 2-multiplicity $1+6+1=8$ and 5-multiplicity $2+2=4$, so $d=8$; likewise in odd degrees. Thus $\lambda_1(M)=b(M)+2(8+8)$. □

The source of the gap is visible in Table 2: in M the dominant prime is 5 in degrees $3,6$ and 2 in degrees $4,5$, so both primes contribute; in C_j' the 2- and 5-torsion occupy the same degrees $3,4$ with equal multiplicity and are absorbed by the maximum. The script `computations/homology/m10_indep.py` reproduces these numbers. It hard-codes placeholder free ranks $b(F)=8$, $b(S)=11$ and prints $\lambda(M)=144$, $\sum\lambda(C_j')=140$, cyclic bound 140; by Corollary 5.4 the gap 4 is independent of these placeholders (also tested for $b(F)\in\{8,22\}$, $b_2(S)\in\{9,30\}$). The actual values are given in Remark 5.5.

Remark 5.5. With $a=19$ (Proposition 4.5) and $\beta=40$ (Proposition 4.9): $b(M)=488$, $\lambda(M)=524$, $\lambda(C_j')=130$, $\#\,\mathrm{Fix}=520$, and the relative deficit is $4/524=1/131$. The value $b(F)=40$ was obtained twice, from two independent constructions of the toric fan (seven rays, nine smooth maximal cones; `computations/sage/bF2.sage`); it depends on the choice of resolution, the gap does not. Over $\mathbf{F}_2$ one has $b(M;\mathbf{F}_2)=b(M)+2\cdot 16=\#\,\mathrm{Fix}$, so the count also equals the total mod-2 Betti number; over $\mathbf{F}_5$, $b(M;\mathbf{F}_5)=b(M)+16<\#\,\mathrm{Fix}$.

5.5. **Proof of Theorem A.** By Theorem 2.3 with $D=8$, each C_j' carries a Morse function h_j with exactly $\lambda(C_j')$ critical points. Proposition 2.6 applied to the Hamiltonian involution $\hat g$ of Proposition 5.3 yields a one-periodic Hamiltonian H on M with $\#\,\mathrm{Fix}(\phi_H^1)=\sum_j\lambda(C_j')$, all nondegenerate, all with contractible orbits since M is simply connected. By Corollary 5.4 this equals $\lambda(M)-4$, and $\lambda(M)=\mathrm{SM}(M)=\mathrm{Morse}(M)$ by Theorem 2.3 with $D=10\geq 9$ ([OpenAI22, Thm. 2.4]). The minimal Chern number is 1 (Proposition 5.3) and $\#\,\mathrm{Fix}=\lambda_1(M)$ by Corollary 5.4. □

Remark 5.6. Theorem A is consistent with Theorem 2.4, which here is an equality. It shows that the cyclic integral Floer bound is sharp in complex dimension 5 (previously complex dimension 1666 [OpenAICompanion]).

6. Dimension eight: necessary conditions and a mechanism obstruction

6.1. Necessary torsion conditions.

Proposition 6.1. *Let M be a simply connected closed symplectic 8-manifold with minimal Chern number N, and put $T_i=\mathrm{Tor}\,H_i(M)$. Then $T_i\cong T_{7-i}$, $T_i=0$ for $i\notin\{2,3,4,5\}$, $\lambda_N(M)=\lambda(M)$ for $N\geq 2$ (including $N=\infty$), and*

$$\lambda(M)-\lambda_1(M)=4\big[d(T_2)+d(T_3)-d(T_2\oplus T_3)\big].$$

Consequently, assuming Theorem 2.4 and Theorem 2.3 for $D=8$, a nondegenerate Hamiltonian on M with fewer than $\mathrm{SM}(M)$ fixed points can exist only if $N=1$ and T_2,T_3 are nonzero with different dominant primes in the sense of (2.1). In particular $\mathrm{Tor}\,H^3(M)\cong T_2\neq 0$.

Proof. $T_i\cong\mathrm{Tor}\,H^{i+1}\cong T_{7-i}$ by universal coefficients and duality; $T_0=T_1=0$ forces $T_6=T_7=0$. If $2N\geq 4$ the degrees $2,3,4,5$ lie in distinct classes mod $2N$. For $N=1$ the even class carries $T_2\oplus T_4\cong T_2\oplus T_3$ and the odd class $T_3\oplus T_5\cong T_3\oplus T_2$; compare with λ, which counts $d(T_2)+d(T_3)$ twice in each parity. The rest follows from Theorems 2.4 and 2.3 and (2.1). □

Remark 6.2. Blowing up a space with free homology along any centers never creates torsion in degree 2 (the shifts in Proposition 2.7 are ≥ 2 and H_0 of a center is free). Heuristically, therefore, a blow-up construction in dimension 8 needs an ambient with Brauer-type torsion from the start; this remark applies only to constructions by blow-ups.

Proposition 6.3 (failure of the product mechanism below dimension 12)**.** *Let X be a simply connected closed Kähler manifold of real dimension ≤ 6 and Y as in §5. Then $\lambda(X\times Y)=4\lambda(X\times\mathbf{CP}^1)$.*

Proof. If $\dim_{\mathbf{R}} X \le 4$, X has no torsion. If $\dim_{\mathbf{R}} X = 6$, the proof of Proposition 3.1 shows that the only torsion is $T := T_2 \cong T_3$. Then $X \times \mathbf{CP}^1$ has torsion T in degrees $2, 3, 4, 5$ and $X \times Y$ has T, T, T^6, T^6, T, T in degrees $2, \dots, 7$, so $\sum t_i(X \times Y) = 16d(T) = 4 \sum t_i(X \times \mathbf{CP}^1)$; free ranks scale by 4 as well. □

6.2. **Perfectness of moment maps over prime fields.**

Lemma 6.4. *Let C be a closed symplectic manifold with a Hamiltonian circle action, moment map μ, and fixed components F of Morse–Bott index $2\mu_F$. For every field k and every i,*

$$\dim H_i(C; k) = \sum_F \dim H_{i-2\mu_F}(F; k).$$

Proof. Let q be a prime. "$\le$": the Morse–Bott inequalities for μ hold with $\mathbf{F}_q$ coefficients, since negative normal bundles are complex, hence oriented, so the Thom isomorphism applies. "$\ge$" for total dimensions: let $X_n = C^{\mathbf{Z}/q^n}$, where $\mathbf{Z}/q^n \subset S^1$. The group $\mathbf{Z}/q^{n+1}$ preserves X_n and acts on it through $\mathbf{Z}/q^{n+1}/\mathbf{Z}/q^n \cong \mathbf{Z}/q$ with fixed set X_{n+1}. The Smith inequality for $\mathbf{Z}/q$ acting smoothly on the closed manifold X_n [Die87, Prop. III.4.16] (stated for smooth $(\mathbf{Z}/q)^r$-manifolds as [Fra16, Prop. 2.1]; see also [Bre72, Ch. III]) gives $\dim H_*(X_{n+1}; \mathbf{F}_q) \le \dim H_*(X_n; \mathbf{F}_q)$. A smooth action of S^1 on a compact manifold has finitely many isotropy groups; since $\bigcup_n \mathbf{Z}/q^n$ is dense in S^1, $X_n = C^{S^1}$ for n large. Hence $\sum_F \dim H_*(F; \mathbf{F}_q) \le \dim H_*(C; \mathbf{F}_q)$. Equality of total dimensions forces the Morse–Bott spectral sequence (with E^1 the shifted homology of the F's) to degenerate, giving degreewise equality. For k of characteristic 0, apply the result for a prime q that divides no torsion order of C or of any F (or use [Fra59; Kir84]). The case of a general field reduces to its prime field. □

Lemma 6.4 combines the classical Smith inequality (see also [AP93, Ch. 3], [Hsi75, Ch. III] for torus actions) with the Morse–Bott inequalities, and we do not claim novelty for it. We could not, however, find an explicit statement in the literature that moment maps of Hamiltonian circle actions are perfect over every $\mathbf{F}_p$. Perfectness is usually stated over $\mathbf{Q}$ or $\mathbf{R}$ [Fra59; Kir84], or over $\mathbf{Z}$ when the fixed points are isolated. The Atiyah–Bott–Kirwan argument via equivariant Euler classes does not apply over $\mathbf{F}_p$ when p divides an isotropy weight, which is why we argue via Smith theory instead.

Corollary 6.5. *In the situation of Lemma 6.4, for every prime p and every i,*

$$r_i(C) = \sum_F r_{i-2\mu_F}(F), \qquad t_i^p(C) = \sum_F t_{i-2\mu_F}^p(F).$$

Proof. $\dim H_i(\,\cdot\,; \mathbf{F}_p) = r_i + t_i^p + t_{i-1}^p$. Subtract the rational identity and induct on i starting from $t_{-1}^p = 0$. □

6.3. **Proof of Theorem B.** Choose a circle $S^1 \subset T$ with $M^{S^1} = M^T$: T has finitely many isotropy subgroups on M, and a circle whose Lie algebra avoids the finitely many proper subspaces $\mathrm{Lie}(H)$, $H \ne T$ an isotropy group, works. Let μ be its moment map. Each component C of $\mathrm{Fix}(g)$ is a T-invariant symplectic submanifold, $\mu|_C$ is a moment map for the circle on C, and $C^{S^1} = C \cap M^T$. Write $\mathcal{F}$ for the set of components of M^T and $\mathcal{F}(C) \subset \mathcal{F}$ for those in C; $\mathcal{F} = \bigsqcup_C \mathcal{F}(C)$. For $F \in \mathcal{F}(C)$ write $2\nu_F$ and $2\mu_F$ for its indices in C and in M.

Step 1: an upper bound for $\lambda(M)$. By Corollary 6.5 applied to M, $b(M) = \sum_F b(F)$ and

$$t_i(M) = \max_p \sum_F t_{i-2\mu_F}^p(F) \le \sum_F \max_p t_{i-2\mu_F}^p(F) = \sum_F t_{i-2\mu_F}(F).$$

Summing over i: $\lambda(M) \le \sum_{F \in \mathcal{F}} \lambda(F)$.

Step 2: equality for each C. We claim $\lambda(C) = \sum_{F \in \mathcal{F}(C)} \lambda(F)$ for every component C of $\mathrm{Fix}(g)$; by hypothesis $\dim_{\mathbf{R}} C \le 6$. (If $\dim_{\mathbf{R}} M = 8$, this holds unless g acts trivially on M, in which case $C = M$ and the theorem is trivial.) Nothing below uses $\dim M$ or $\pi_1(C)$. A

component F with nonzero torsion has $\dim_{\mathbf{R}} F \geq 4$ (closed orientable manifolds of dimension ≤ 2 have free homology). If $F = C$ there is nothing to prove. Otherwise $\dim_{\mathbf{R}} C = 6$ and $\dim_{\mathbf{R}} F = 4$, and the normal bundle of F in C is a complex line, so $\nu_F \in \{0, 1\}$ and F is a local minimum or local maximum of $\mu|_C$. By the connectedness theorem of Atiyah and Guillemin–Sternberg [Ati82; GS82], local extrema are global and the minimum and maximum components are unique. Hence C contains at most two torsion-carrying fixed components: the minimum F_- ($\nu = 0$, torsion in degrees $1, 2$) and the maximum F_+ ($\nu = 1$, torsion in degrees $3, 4$). These degree supports are disjoint, so by Corollary 6.5 each $t_i(C)$ equals $t_i(F_-)$ or $t_{i-2}(F_+)$ or 0, whence $\sum_i t_i(C) = \sum_{F \in \mathcal{F}(C)} \sum_i t_i(F)$. Together with $b(C) = \sum_{F \in \mathcal{F}(C)} b(F)$ this proves the claim.

Step 3. $\sum_j \lambda(C_j) = \sum_j \sum_{F \in \mathcal{F}(C_j)} \lambda(F) = \sum_{F \in \mathcal{F}} \lambda(F) \geq \lambda(M)$.

Consequence. The fixed points of $\psi^\epsilon \circ g$ are the critical points of Morse functions h_j on C_j, and $\#\operatorname{Crit}(h_j) \geq \lambda(C_j)$ by Proposition 2.2. Hence $\#\operatorname{Fix} \geq \lambda(M)$, which equals $\mathrm{SM}(M)$ for simply connected M by Theorem 2.3 ($D = 8$). □

Remark 6.6 (a support criterion). The proof shows, in any dimension, that $\sum_j \lambda(C_j) \geq \lambda(M)$ whenever, in every C_j, the index-shifted torsion supports of the torsion-carrying fixed components are pairwise disjoint. A construction of this type in dimension $2n$ therefore needs a component C_j with two torsion-carrying T-fixed components whose shifted supports overlap. In Theorem A this happens as follows. The diagonal circle has weights $(1, 1)$ at $(0, 0)$ and $(-1, -1)$ at (∞, ∞), so it acts *trivially* on E_{00} and $E_{\infty\infty}$; hence C_1' and C_4' (over these curves) are themselves T-fixed components. On C_2' and C_3' (over $E_{0\infty}$, $E_{\infty 0}$) the circle rotates the $\mathbf{CP}^1$-factor, and the T-fixed components are three: the proper transform of $F \times \{y_j\}$ ($\cong F$, since $S \subset F$ is a divisor), $F \times \{y_j'\}$ for the other fixed point y_j' of E_j, and the section $\mathbf{P}(T_{y_j} E_j) \cong S$ of the exceptional divisor (the section $\mathbf{P}(N_{S/F})$ lies in the proper transform of $F \times \{y_j\}$). As a check, the total Betti number of M^T is $2 \cdot 122 + 2\,(40 + 40 + 42) = 488 = b(M)$, as Lemma 6.4 requires. Within C_2' one has $\sum_F \lambda(F) = 44 + 44 + 46 = 134 > 130 = \lambda(C_2')$: the loss of 4 comes from the overlap of the 5-torsion of S (degrees $3, 4$ after the index shift) with the 2-torsion of the copies of F (degrees 2–5). Altogether $\sum_{F \subset M^T} \lambda(F) = 528 \geq \lambda(M) = 524 \geq \sum_j \lambda(C_j') = 520$, consistent with Step 1.

Remark 6.7 (a second argument via Li's theorem). By a theorem of H. Li [Li03, Thm. 0.1], for a Hamiltonian circle action on a compact symplectic manifold C, $\pi_1(C) \cong \pi_1(F_{\min}) \cong \pi_1(F_{\max}) \cong \pi_1(C_{\mathrm{red}})$. If $\dim C = 6$ and C is simply connected, its four-dimensional fixed components (necessarily extremal) are simply connected, hence torsion-free, and the torsion-carrying fixed components of M are themselves components of $\operatorname{Fix}(g)$; the inequality of Step 1 then gives the claim. If some C_j is not simply connected, Lemma 6.4 prevents cancellation of its H_2-torsion. This gives an alternative route to Theorem B; we have not written it out in full.

6.4. **Hamiltonian maps of prime order outside torus actions.** Theorem B concerns finite-order elements of Hamiltonian torus actions. For an arbitrary Hamiltonian diffeomorphism of prime order we still obtain a strong necessary condition, using a recent vanishing theorem of Bai–Shelukhin–Wilkins–Xu.

Proposition 6.8. *Let (M, ω) be a closed simply connected symplectic manifold of real dimension 8, and let $g \in \operatorname{Ham}(M, \omega)$ have prime order p'. Write $\operatorname{Fix}(g) = C_1 \sqcup \cdots \sqcup C_r$, put $T_i = \operatorname{Tor} H_i(M; \mathbf{Z})$, and suppose that*

$$\sum_j \lambda(C_j) < \lambda(M),$$

i.e. that perturbing g as in Proposition 2.6 with Morse functions attaining $\lambda(C_j)$ would beat $\lambda(M)$. Let P, Q be primes with $d(T_2) = \operatorname{rank}_P T_2$ and $d(T_3) = \operatorname{rank}_Q T_3$. Then $P \neq Q$, and some component C_j has, in a single degree, a torsion subgroup containing $\mathbf{Z}/PQ$. More precisely, either C_j is a simply connected 6-manifold with $\mathbf{Z}/PQ \subset \operatorname{Tor} H_2(C_j)$, or C_j is not

simply connected and $\mathbf{Z}/PQ$ *embeds in* $\operatorname{Tor} H_1(C_j)$ *or* $\operatorname{Tor} H_2(C_j)$*. In particular* g *is not a finite-order element of any Hamiltonian torus action (Theorem B).*

Proof. Step 1: Betti-number relations.

- By [Bai+26, Prop. C.17(1)], for every odd prime $p \neq p'$ the Morse–Bott Floer complex of g with coefficients in the Novikov field over $\mathbf{F}_p$ has vanishing differential. Hence $b(M;\mathbf{F}_p) = \sum_j b(C_j;\mathbf{F}_p)$ for all such p. (The local Floer homology of a fixed component of a finite-order map is its ordinary homology; [Bai+26, Prop. C.9].)
- Taking p larger than all torsion orders of M and of the C_j gives $b(M;\mathbf{Q}) = \sum_j b(C_j;\mathbf{Q})$.
- For every prime p, the Morse–Bott spectral sequence gives $b(M;\mathbf{F}_p) \leq \sum_j b(C_j;\mathbf{F}_p)$.

Writing $b(X;\mathbf{F}_p) = b(X;\mathbf{Q}) + 2\sum_i t_i^p(X)$, where $t_i^p(X)$ is the p-rank of $\operatorname{Tor} H_i(X)$, we obtain for every prime p

$$\sum_j \rho_p(C_j) \;\geq\; \operatorname{rank}_p T_2 + \operatorname{rank}_p T_3, \qquad \rho_p(X) := \tfrac{1}{2}\textstyle\sum_i t_i^p(X). \tag{6.1}$$

Here we used $\sum_i t_i^p(M) = 2(\operatorname{rank}_p T_2 + \operatorname{rank}_p T_3)$, by Poincaré duality and simple connectivity.

Step 2: the torsion part of λ *for each component.* For each component, $\lambda(C_j) - b(C_j) = 2\sum_i d(\operatorname{Tor} H_i C_j) =: 2\tau_j$.

- If C_j is a simply connected 6-manifold, then $\tau_j = 2\,d(\operatorname{Tor} H_2 C_j)$ and $\rho_p(C_j) = \operatorname{rank}_p \operatorname{Tor} H_2 C_j$.
- For a 4-manifold, $\tau_j = 2\,d(\operatorname{Tor} H_1 C_j)$ and $\rho_p = \operatorname{rank}_p \operatorname{Tor} H_1 C_j$.
- For a non-simply-connected 6-manifold, $\tau_j = 2\,(d(\operatorname{Tor} H_1) + d(\operatorname{Tor} H_2))$ and $\rho_p = \operatorname{rank}_p \operatorname{Tor} H_1 + \operatorname{rank}_p \operatorname{Tor} H_2$.
- Components of dimension ≤ 2 are torsion-free.

In all cases $\tau_j \geq 2\max_p \rho_p(C_j)$. Equality with $2(\rho_P + \rho_Q)$ fails only if P- and Q-torsion coexist in a single degree.

Step 3: conclusion. The rational parts cancel by Step 1, so the hypothesis says $\sum_j \tau_j < 2(d(T_2) + d(T_3)) = 2(\operatorname{rank}_P T_2 + \operatorname{rank}_Q T_3)$.

- If $P = Q$, then (6.1) gives $\sum_j \tau_j \geq 2\sum_j \rho_P(C_j) \geq 2(\operatorname{rank}_P T_2 + \operatorname{rank}_P T_3)$, a contradiction.
- If $P \neq Q$ and no component mixes P- and Q-torsion in a single degree, then $\tau_j \geq 2(\rho_P(C_j) + \rho_Q(C_j))$ for every j. Summing and using (6.1) for P and for Q gives $\sum_j \tau_j \geq 2(\operatorname{rank}_P T_2 + \operatorname{rank}_P T_3 + \operatorname{rank}_Q T_2 + \operatorname{rank}_Q T_3)$, again a contradiction.

Hence some component mixes P- and Q-torsion in one degree, which is the stated alternative. The last sentence follows from Theorem B. □

Remark 6.9. For $p = p'$ one also has equality in (6.1), by the classical Smith inequality. Atallah–Shelukhin [AS20, Thm. G] had earlier obtained similar statements in the monotone case.

Hamiltonian finite-order maps that do not embed in a Hamiltonian circle action are known only in real dimension 4. Chiang–Kessler [CK19, §5] give an involution of $\mathbf{CP}^2\#6\overline{\mathbf{CP}^2}$; that it is Hamiltonian is proved in [LLW22, Cor. 1.8]. Whether such maps exist in higher dimensions is open [LLW22, Question 1.10].

6.5. **Dimensional limitation of transverse centers (heuristic).** The following is a heuristic, not a theorem. In Theorem A the extra torsion of the fixed components is introduced by a blow-up whose center W meets $\operatorname{Fix}(g)$ cleanly and satisfies (5.1). If the inserted torsion is that of a surface (S, real dimension 4, with odd-prime torsion in H_1), then W must contain S and, by (5.1), the normal directions $E^{\neq}$ of the fixed component (real dimension ≥ 2), and must have real codimension ≥ 4 (a divisor blow-up is the identity). Hence $\dim_{\mathbf{R}} M \geq 4 + 2 + 4 = 10$. The argument assumes that the inserted torsion comes from a surface contained in the center; it is independent of, and weaker than, Theorem B.

7. Higher dimensions, earlier claims, and open problems

7.1. Kähler examples in dimensions twelve and sixteen.

Remark 7.1 (dimension 12). Let $X^8 = \mathrm{Bl}_{S\times\{t_1\}\sqcup S\times\{t_2\}}(F \times \mathbf{CP}^1)$ and $M^{12} = X^8 \times Y$, $C = X^8 \times \mathbf{CP}^1$. Then M^{12} is simply connected Kähler, and the product mechanism gives $\lambda(M^{12}) - 4\lambda(C) = 8$; with $b(F) = 40$, $b(S) = 42$ one gets $\lambda(M^{12}) = 1400$ and $\#\,\mathrm{Fix} = 1392$. The gap relies on Lemma 4.6 (no 5-torsion in F); with 5-torsion of rank ≥ 1 in F it would vanish.

Remark 7.2 (dimension 16). In $P = \mathbf{CP}^1 \times \mathbf{CP}^5$ embed disjointly (by [OpenAI22, Lemma 3.2]) a bielliptic surface S_2 with $(\mathbf{Z}/2)^2$ torsion and three bielliptic surfaces $S_3^{(i)}$ with $\mathbf{Z}/3$ torsion into $\mathbf{CP}^5$, and blow up P along $S_2 \times \mathbf{CP}^1$ (codimension 3) and $S_3^{(i)} \times \{t\}$ (codimension 4). For the resulting complex sixfold X and $M = X \times Y$ one finds $\lambda(M) = 1232$, $4\lambda(X \times \mathbf{CP}^1) = 1216$ (two independent scripts, `computations/homology/indep.py` and `computations/homology/x12.py`); the family with $3m$ copies of S_3 and m copies of $S_2 \times \mathbf{CP}^1$ has $\lambda = 96 + 1136m$ and gap $16m$, so the relative deficit is $16m/(96 + 1136m) \geq 1/77$, tending to $1/71$ as $m \to \infty$. Here $\#\,\mathrm{Fix} = 1216$ also equals the Bai–Xu cyclic bound $\lambda_1(M)$ (minimal Chern number 1), as recomputed with `computations/homology/indep.py`. This is a rearrangement of the centers of [OpenAI22] and is not claimed as a separate result.

Remark 7.3 (limits of surface centers in free ambients). Suppose X is obtained from a simply connected compact Kähler manifold with torsion-free homology by blowing up disjoint centers of the form $S \times B$, where S is a surface and B has torsion-free homology concentrated in even degrees. This is the method of [OpenAI22]. Put $n = \dim_{\mathbf{C}} X$.

By Proposition 2.7, a center $S \times B$ of complex codimension c contributes its torsion in each parity list with profile $e_B(u)\,(u + \cdots + u^{c-1})$, where e_B is the even Poincaré polynomial of B in $u = t^2$.

- For $n = 5$ the only possibilities are $B = \mathrm{pt}$ with $c = 3$ and $B = \mathbf{CP}^1$ with $c = 2$. Both give the profile $u + u^2$.
- For $n = 4$ the only center is $B = \mathrm{pt}$ with $c = 2$. Its torsion lies in degrees $3, 4$, and none lies in degree 2.

In either case the profiles of different primes are proportional or occupy a single pair of dual degrees. A direct computation (same scripts) then gives $\lambda(X \times Y) = 4\lambda(X \times \mathbf{CP}^1)$ (`computations/homology/surface_limit.py`).

Hence the product construction $X \times Y$ of [OpenAI22], with such X, produces no deficit below real dimension 16. Going below this requires torsion of a different shape. In Theorem A this is the Brauer torsion of F together with a center that is not of product type.

7.2. Relation to an earlier claim.

For a closed symplectic manifold (M, ω), Ma [Ma13] writes $\mathrm{Arn}(M, \omega)$ for the minimal number of fixed points of a Hamiltonian symplectomorphism of M and $\mathrm{Arn}_s(M, \omega)$ for the minimum over those all of whose fixed points are nondegenerate. He writes $\mathrm{Crit}(M)$ for the minimal number of critical points of a smooth function and $\mathrm{Crit}_m(M)$ for the minimal number of critical points of a Morse function. His Theorem 1.5 reads: "Let (M, ω) be a close[d] symplectic manifold with symplectic form ω. Then $\mathrm{Arn}(M, \omega) = \mathrm{Crit}(M)$ and $\mathrm{Arn}_s(M, \omega) = \mathrm{Crit}_m(M)$" [Ma13, Thm. 1.5]. There are no further hypotheses, and all fixed points are counted, not only those with contractible orbits. See also [Vib], cited for context only.

The second equality is already refuted by [OpenAISept23, Thm. 1.1], on a closed symplectic manifold of real dimension 12 with $\pi_1 = A_5^k$, which is not simply connected, and by the simply connected examples of [OpenAIStable; OpenAI22; OpenAICompanion] in higher dimensions. The first equality is refuted by the degenerate example of [OpenAIDeg] on the quadric Q^3. For the manifold M of Theorem A, $\mathrm{Crit}_m(M) = \mathrm{Morse}(M) = \mathrm{SM}(M) = \lambda(M)$ by Theorem 2.3, while $\#\,\mathrm{Fix}(\phi_H^1) = \lambda(M) - 4$, so $\mathrm{Arn}_s(M, \omega) \leq \mathrm{Crit}_m(M) - 4$. Theorem A

is thus the first simply connected example below the stable Morse number in real dimension less than 22, and it refutes the second assertion of [Ma13, Thm. 1.5] on a simply connected Kähler manifold of real dimension 10.

7.3. **Open problem: dimension eight.** By Theorems A and B and Proposition 3.1, among simply connected closed symplectic manifolds of real dimension at least 6 the minimal real dimension of a counterexample to (1.1) is 8 or 10 (Corollary 1.1). Simply connected real dimension 4 remains open in the sense of §3, and non-simply-connected manifolds of real dimension 4, 6, 8, 10 are not addressed here.

Problem 7.4. Does there exist a closed (simply connected, Kähler) symplectic 8-manifold M and a nondegenerate Hamiltonian diffeomorphism with fewer than $\mathrm{SM}(M)$ fixed points?

By Proposition 6.1, such an M must have $N = 1$ and Brauer-type torsion $T_2 \neq 0$ whose dominant primes differ from those of T_3. By Theorem B, the Hamiltonian cannot be obtained by perturbing a finite-order element of a Hamiltonian torus action. A positive answer requires either a finite-order Hamiltonian diffeomorphism not contained in any Hamiltonian torus action, whose fixed set contains a component with $\mathbf{Z}/PQ$-torsion in a single degree (Proposition 6.8), or an entirely different mechanism. A negative answer requires a Floer-theoretic obstruction stronger than Theorem 2.4, which by Proposition 6.1 is not sufficient in general.

Appendix A. Computational corroboration

The script `computations/milp/lpsearch2.py` formulates a mixed-integer linear program modelling the torsion bookkeeping of the finite-order/torus mechanism in real dimension $2n$. Variables are nonnegative weights x of "torsion pieces" (prime $\in \{p, q\}, f, (a, b), c, j, \nu, \varepsilon$): a pair of Poincaré-dual cyclic summands in degrees $a < b$ of a T-fixed component of complex dimension f, contained in a component C of $\mathrm{Fix}(g)$ of complex dimension c with label $j \leq K$, with index 2ν in C and $2(\nu + \varepsilon)$ in M. Constraints: Poincaré duality of the torsion profiles of M and of each C; no torsion of M in degrees $0, 1$; uniqueness of minimum and maximum components in each C and globally; if $F = C$ then C contains no other piece; total weight ≤ 1. The objective is the normalized gain $\sum_k \max(\mathrm{profile}_M(k)) - \sum_C \sum_k \max(\mathrm{profile}_C(k))$, linearized with binary selectors. The modelling of the torsion of C as the sum of the shifted torsion of its fixed components is justified by Corollary 6.5, so no extra perfectness hypothesis is needed.

real dim	setting	max normalized gain
8	$K = 2, 3$; with or without H_1-torsion in C	0
10	$K = 2, 3$, no H_1-torsion in C	1/3
10	$K = 3$ unrestricted	1/2
12	$K = 2, 3$	1 (positive control)
10	forbid 6-dim F / Brauer-type $(2, 3)$ F / surfaces / 8-dim C	0 in each case

Table 4. Results (`computations/milp/run_lp.py`, `computations/milp/necessity.py`).

In dimension 10 the optimal configurations contain an 8-dimensional C whose minimum and maximum (or saddles) are 6-dimensional Brauer-type components and which contains torsion surfaces of the other prime at index 2; by Guillemin–Sternberg birational cobordism [GS89], such a surface is a codimension-2 symplectic submanifold of a reduced space of Brauer type, which led to the construction of §4. *Caveats:* this is an LP relaxation with real weights and two primes, it encodes only the listed necessary conditions and not realizability, and it is used only as corroboration; Theorem B has a complete hand proof.

Appendix B. Machine verification of the symmetroid and the quintic

All paths are relative to the repository root. In exact arithmetic over $\mathbf{Q}$ (and over $\overline{\mathbf{Q}}$ where stated):

- `computations/sage/q10.sage` (random seed 7) produces the web and the quintic and verifies: the character $f(gx) = \zeta^3 f(x)$; irreducibility of f over $\mathbf{Q}$; that the saturated Jacobian ideal of f has Hilbert polynomial 10 and equals its radical and the saturated ideal of 3×3 minors of A; that the coordinate points lie on B and are smooth points (gradients $(0, 0, -1728, 0)$, $(0, 0, 0, -875)$, $(0, -640, 0, 0)$, $(-672, 0, 0, 0)$ at $p_0, \dots, p_3$); that the web has no base points; that a general (non-invariant) web also has total Tjurina number 10; that the invariant quintic Q is smooth, avoids the coordinate points and the nodes, and that $B \cap \{Q = 0\}$ is a smooth curve (Hilbert polynomial $20t - 50$: degree 20, genus 51).
- `computations/sage/beauville_ii.sage`: the rank-≤ 1 locus of the web is empty, and Beauville's condition (ii) holds at each of the ten rank-2 points, computed exactly over $\overline{\mathbf{Q}}$.
- `computations/sage/defect2.sage`: the ideal of the nodes is radical of degree 10, and its Hilbert function is $1, 4, 10, 10, \dots$, so the nodes impose independent conditions on quadrics.
- `computations/sage/bF2.sage`: the toric resolution of $\frac{1}{5}(1, 1, 4)$ by iterated star subdivision has 7 rays and 9 smooth maximal cones; an independent construction of the fan gives the same result.
- `computations/sage/symm.sage`: auxiliary experiments with random invariant webs.

The homology bookkeeping of §5 is implemented twice, with identical results: `computations/homology/lam.py` together with `w10.py` tracks torsion profiles, and `computations/homology/indep.py` together with `m10_indep.py` tracks degreewise integral groups with Künneth. `computations/homology/x12.py` contains the higher-dimensional scans. The MILP of Appendix A is `computations/milp/lpsearch2.py`, run by `computations/milp/run_lp.py` and `computations/milp/necessity.py`.

Statement on the use of AI

The results in this manuscript were found, checked and written up by Claude (Anthropic) agents working under the direction of the author. The computations listed above were run by those agents. The manuscript has not undergone human peer review.

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